

When ideas are difficult to explain, I build them.

Creative Technologist and Product Designer leveraging AI, code, design systems, and cognitive psychology to transform emerging concepts into tangible product experiences.

CONTACT

Bradenton, FL
melissa.casole@yahoo.com
618-795-4580
www.melissacasole.com
Portfolio access: **MelissaAI123!**
linkedin.com/in/melissacasole

FOCUS

- AI Product Design
- Interactive Prototyping
- Design Systems as Platforms
- Hardware + Software UX
- Healthcare & Med-Device
- Cognitive Psychology

TECHNOLOGY

AI

Claude Code · Cursor · ChatGPT ·
Gemini · ElevenLabs

DESIGN

Figma · ProtoPie · FigJam

DEVELOPMENT

Vercel · GitLab · HTML · CSS ·
JavaScript · Arduino C++

SYSTEMS

Design Tokens · Storybook ·
Component Architecture

PROFILE

I reduce uncertainty by building realistic experiences stakeholders can see, test, and understand — working software over static mockups, demonstrations over decks. My work spans AI-powered clinical platforms, physical hardware prototypes, design systems built as product platforms, and founder-built tools. I don't wait for the future to arrive. I prototype it.

FEATURED WORK

Arti Wall

AI-POWERED OPERATING ROOM INTELLIGENCE

Transformed a leadership vision into a living product experience: an ambient OR intelligence platform with LLM-powered voice interaction, designed around four clinical personas and demonstrated to executives, engineers, and surgeons.
→ Funded roadmap initiative targeting MVP release in 2027.

Horizon Control

HARDWARE + SOFTWARE INTERACTION DESIGN

Screen prototypes couldn't simulate surgery — so I built a 3D-printed surgical camera wired to Arduino with AI-generated C++, driving tablet and HUD interfaces in real time through ProtoPie Connect for surgeon testing.
→ Validated four interaction models; influenced future Vision platform direction.

Vision Design System

DESIGN SYSTEMS AS PRODUCT PLATFORMS

Unified independently evolving surgical products (CCU, Trident, GX) under one foundation: Figma Variables and design tokens migrated into a Vuetify implementation, with an interactive documentation platform covering motion, elevation, and behavior.
→ Scalable foundation and governance model for future Vision products.

ProtoCue · Founder

BUILDING THE TOOL I NEEDED

AI-generated coded prototypes broke traditional review workflows, so I designed, built, and shipped the solution: contextual feedback attached directly to live interactive experiences. Built end-to-end with Claude Code, Cursor, GitLab, and Vercel.
→ Working product, actively evolving for AI-first product teams.

INDUSTRIES

- Healthcare AI
- Medical Devices
- Enterprise SaaS
- Retail

EDUCATION

B.S. Computer Science & Design

Missouri College, 2007

Minor: Cognitive Psychology

CERTIFICATIONS

- Human Factors for Medical Devices — AAMI, 2024
- UX Design Certification — Google, 2020
- Human-Computer Interaction — MIT, 2019

INDEPENDENT PRODUCTS

PROTOCUE

Contextual feedback on live interactive experiences.

AI MEETING INTELLIGENCE

Captures meetings, summarizes, extracts action items.

KEY ACHIEVEMENT

Code-First Product Validation

Pulled production design-system components into AI-assisted development workflows, letting teams validate production-accurate experiences before engineering investment.

Figma → Claude Code → Production Components → Vercel → MS Clarity / Hotjar → User Testing → Engineering

PROFESSIONAL EXPERIENCE

Creative Technologist · Product Design Consultant

2023 – Present

Arthrex

- Transformed leadership's operating-room vision into Arti Wall, an AI voice + LLM product experience that aligned executives, engineers, PMs, and clinicians — now a funded roadmap initiative (MVP 2027).
- Built Horizon Control, a physical Arduino prototype with AI-generated C++ and ProtoPie Connect, validating four interaction models in surgeon testing.
- Created the Vision Design System — Figma Variables, design tokens, and a Vuetify implementation — as the platform for future Vision products.

Sr Designer & UX Manager

Sept 2020 – Present

First Advantage

- Created and evolved the ELEMENTS design system (v2 → v3): 150+ components aligned across React, Angular, Vue, and Web — 30% less UI debt, 40% less design-to-dev rework.
- Pioneered code-first product validation: production components in AI-assisted workflows, validated on Vercel with MS Clarity/Hotjar before engineering investment.
- Increased sprint velocity 25% with AI-assisted prototyping; raised accessibility scores 78% → 96%; mentored a UX team of 6 with 100% retention.

Contract Senior UX Designer

Mar 2020 – Sept 2020

Motivus (formerly AgileThought)

Figma/ProtoPie prototypes cut stakeholder approval cycles 35%; journey mapping drove a redesign reducing task time 22%.

UX Engineer

Oct 2018 – Mar 2020

Mediagistic

Designed an AI/ML-driven ad delivery platform (+18% CTR, -12% spend waste); built front-end prototypes saving 10+ dev hours per sprint.

Senior UX Designer

May 2017 – Oct 2018

Inspirata

Co-designed digital cancer diagnostics — 30% faster pathologist workflows; usability testing with 50+ pathologists informed FDA submission readiness.

Lead UX Designer

Jun 2016 – May 2017

MarineMax

Created the company's first design system; POS software adopted by 100+ retail locations.